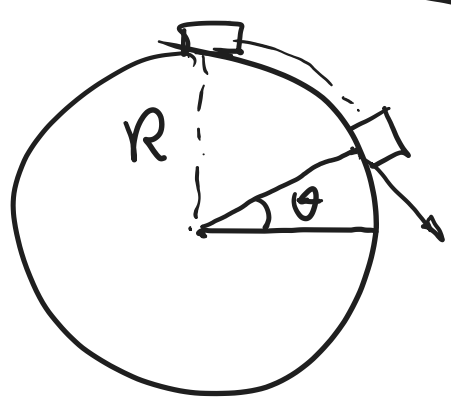


Problem 1) The work-energy theorem states that the initial release velocity would be: $v_0 = \sqrt{2gl \cos \theta}$ where l represents the string length. For projectile motion, the horizontal distance traveled is

$$x = \frac{2v_0^2 \sin \theta \cos \theta}{g} = \frac{2(2gl \cos \theta) \sin \theta \cos \theta}{g}$$

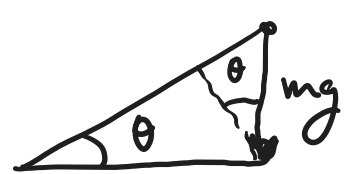
$$\frac{dx}{d\theta} = 0 \Rightarrow \cos \theta - 2 \cos \theta \sin^2 \theta = 0 \Rightarrow \cos^2 \theta = 2 \sin^2 \theta \Rightarrow \boxed{\tan \theta = \frac{1}{\sqrt{2}}}$$

Problem 2) The detachment occurs where the inward component of gravitational force becomes insufficient to represent the centripetal force. Let this point's angle be θ .



$$\Rightarrow F_c = \frac{mv^2}{R} = mg \sin \theta \quad \text{①} \quad \text{two unknowns. But we also}$$

know from work-energy theorem that the vertical distance from the top to the detachment point is $R - R \sin \theta$.



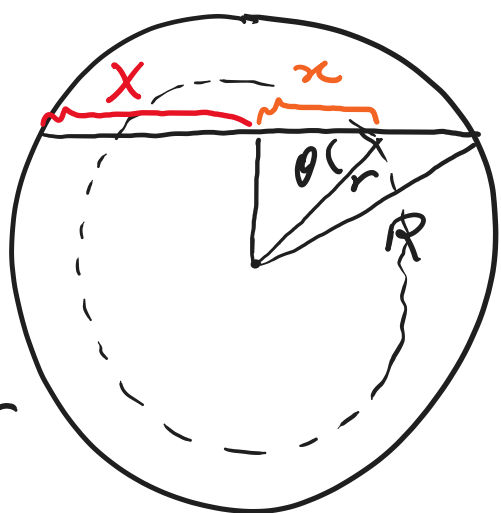
$$\left. \begin{aligned} \frac{1}{2}mv^2 &= mgR(1 - \sin \theta) \quad \text{②} \\ \text{①} \end{aligned} \right\} \Rightarrow 2mg(1 - \sin \theta) &= mg \sin \theta$$

$$\Rightarrow 3 \sin \theta = 2 \Rightarrow \sin \theta = \frac{2}{3} \Rightarrow \boxed{\theta \approx 0.7297 \text{ rad}}$$

Problem 3) 1) Where the ball reaches the top of earth center.

2) The gravitational force at radius r due to mass M_r inside a sphere of radius $r \leq R$ is,

$$F_g = \frac{GM_r m}{r^2} = \frac{Gm(4\pi r^3 \rho / 3)}{r^2} = \frac{4}{3} \pi G m \rho r$$



$$3) F = mv \frac{dv}{dx} = -F_g \cos \theta = -\frac{4}{3} \pi G m \rho x$$

$$\Rightarrow \int_0^v mv dv = - \int_x^0 \frac{4}{3} \pi G m \rho x dx \Rightarrow \frac{1}{2}mv^2 = \frac{4}{3} \pi G m \rho X$$

$$\Rightarrow \boxed{v = \sqrt{\frac{8}{3} \pi G \rho X}}$$

Problem 4) The differential potential is $d\phi = -G \frac{dM}{r}$ where $dM = \rho dA = \rho 2\pi x dx$.

$$\Rightarrow d\phi = -2\pi \rho G \frac{x dx}{r} = -2\pi \rho G \frac{x dx}{r} = -2\pi \rho G \frac{x dx}{(x^2 + z^2)^{1/2}}$$

$$\Rightarrow \phi(z) = -2\pi \rho G \int_0^a \frac{x dx}{(x^2 + z^2)^{1/2}} = -2\pi \rho G (x^2 + z^2)^{1/2} \Big|_0^a$$

$$= -2\pi \rho G [(a^2 + z^2)^{1/2} - z]$$

$F = -\nabla U = -m \nabla \phi$, problem is symmetric, therefore,

$$\boxed{F_z = -m \frac{\partial \phi}{\partial z} = 2\pi m \rho G \left[\frac{z}{(a^2 + z^2)^{1/2}} - 1 \right]}$$

Problem 5) The expected orbital speed v due to the galaxy mass M within radius R is,

$$\frac{GM}{r^2} m = \frac{mv^2}{R}$$

$$\text{where } m \text{ is the orbital mass. } \Rightarrow v = \sqrt{\frac{GM}{R}}$$

Because the luminous galactic mass ends beyond the galactic limbo, we would expect $v \propto \sqrt{\frac{1}{R}}$ when $R > \text{limb length}$.

However, we observe $v \propto \text{const.}$ which implies $\boxed{M(R) \propto R}$